



DATEV CODING
DAY

Advanced Querying Techniques With Aggregation Pipelines



Tobias Joschko
Solutions Architect



Hands-on session

- No stupid questions
- Actively discuss
- Be respectful





Hi, I'm Tobias Joschko!

You can find me on LinkedIn.



**Tobias
Joschko**

- Solutions Architect at MongoDB
- Software Developer
- Lecturer for Software Development

Agenda

 MongoDB Query API

 Common Aggregation Stages

 Building & Running Pipelines

 Wrap Up

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 **MongoDB Query API**

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MongoDB Query API

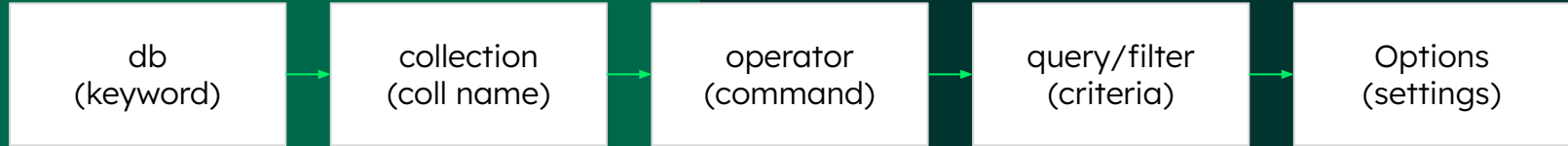
CRUD



That is a bunch of crud.



MongoDB Query API Syntax Structure



// Example MongoDB find command

```
db.collection.find(  
  { age: { $gte: 25 } }, // query/filter: Find documents where age is greater than or equal to 25  
  {  
    projection: { name: 1, age: 1 }, // options/settings: Include only the name and age fields in the  
    results  
    sort: { age: -1 },           // options/settings: Sort the results by age in descending order  
    limit: 10                   // options/settings: Limit the number of results to 10  
  }  
);
```



Create

```
db.authors.insertOne({  
    "name": "Jane Austen",  
    "books": [ "0141439688", "0375757813", "1551114798" ],  
    "aliases": [ "Austen, Jane", "Jane Austen" ]  
})
```



Read

```
db.authors.find({  
  "name": "Jane Austen"  
})
```



Update

```
db.authors.updateOne(  
  { name: "Jane Austin" },  
  { $set:  
    { name: "Jane Austen" }  
  }  
)
```



Delete

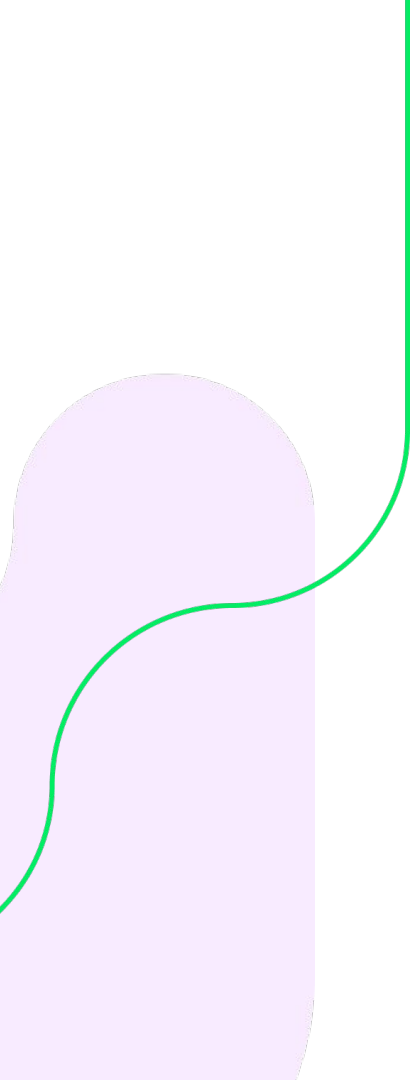
```
db.authors.deleteOne(  
    { name: "Jane Austen" }  
)
```

Check out the
MongoDB docs
for more
details



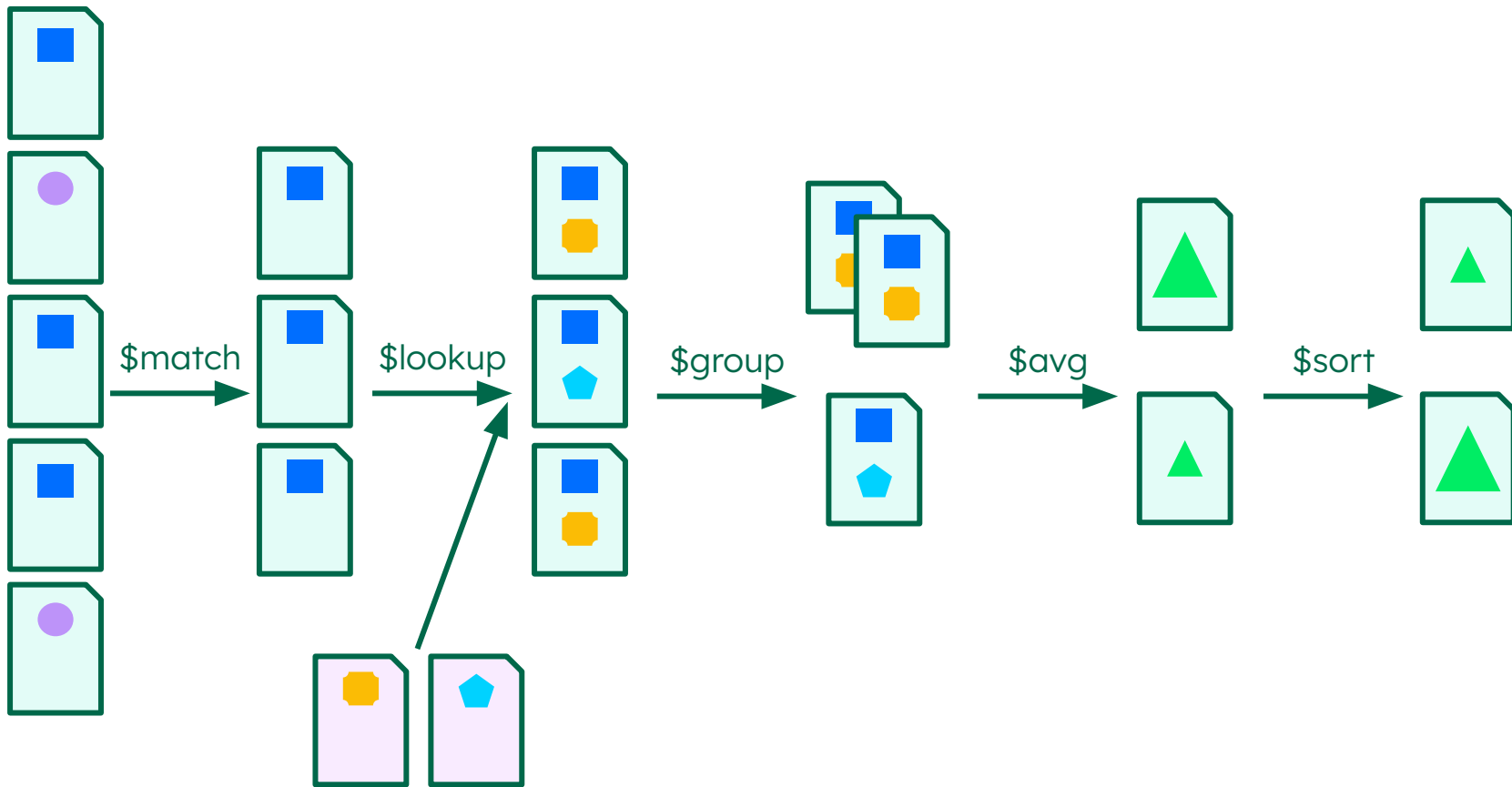
Aggregation Framework





The **aggregation
framework** allows
you to transform
and analyze your
data in real time.

Multi-Stage Aggregation Pipelines





Look familiar?

`ps aux | grep python | sort -nrk 3 | head -5`

```
michael.lynn@M-JG30DWXC36 chatbot-vector-search % ps aux
```

```
USER      PID %CPU %MEM    VSZ   RSS  TT  STAT STARTED   TIME COMMAND
```

```
michael.lynn  47511 41.6 0.5 1598815648 171952 ?? S   9:44AM  1:34.69 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/126.0.64
```

```
_windowserver  378 15.8 0.6 418242064 211136 ?? Ss  9Jul24 1581:30.91 /System/Library/PrivateFrameworks/SkyLight.framework/Resources/WindowServer -daemon
```

```
michael.lynn  842 10.7 0.3 419056160 98096 ?? S   9Jul24 145:58.49 /Applications/iTerm.app/Contents/MacOS/iTerm2
```

```
michael.lynn  808 5.7 3.0 447883984 1131872 ?? S   9Jul24 344:35.07 /Applications/Google Chrome.app/Contents/MacOS/Google Chrome
```

```
michael.lynn  42220 5.7 1.6 1610563504 619008 ?? S   9:39AM  0:42.23 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/126.0.64
```

```
michael.lynn  62628 3.0 0.4 413846304 158720 s058 S+  7:42PM  9:18.10 /Library/Frameworks/Python.framework/Versions/3.12/Resources/Python.app/Contents/MacOS/Python app.py
```

```
root          627 2.2 0.4 412929632 136896 ?? Ss  9Jul24 539:58.74 /Library/SystemExtensions/BBFBE1C6-5010-43FA-A744-9141EE144E41/com.code42.agent.extension.systemextensi
```

```
michael.lynn  959 1.7 1.1 1598913904 397232 ?? S   9Jul24 116:01.35 /Applications/Slack.app/Contents/Frameworks/Slack Helper (Renderer).app/Contents/MacOS/Slack Helper (Re
```

```
michael.lynn  763 1.7 0.2 413714448 58992 ?? S   9Jul24 45:43.74
```

```
/System/Library/PrivateFrameworks/UniversalAccess.framework/Versions/A/Resources/AXVisualSupportAgent.a
```

```
_mysql        51517 1.7 1.5 412731408 562720 ?? Rs  9:48AM  0:01.07 /usr/local/mysql/bin/mysqld --user=_mysql --basedir=/usr/local/mysql --datadir=/usr/local/mysql/data --
```

```
root          613 1.2 0.2 412605120 59904 ?? Ss  9Jul24 114:26.79 /Library/Application Support/JAMF/Jamf.app/Contents/MacOS/JamfDaemon.app/Contents/MacOS/JamfDaemon
```

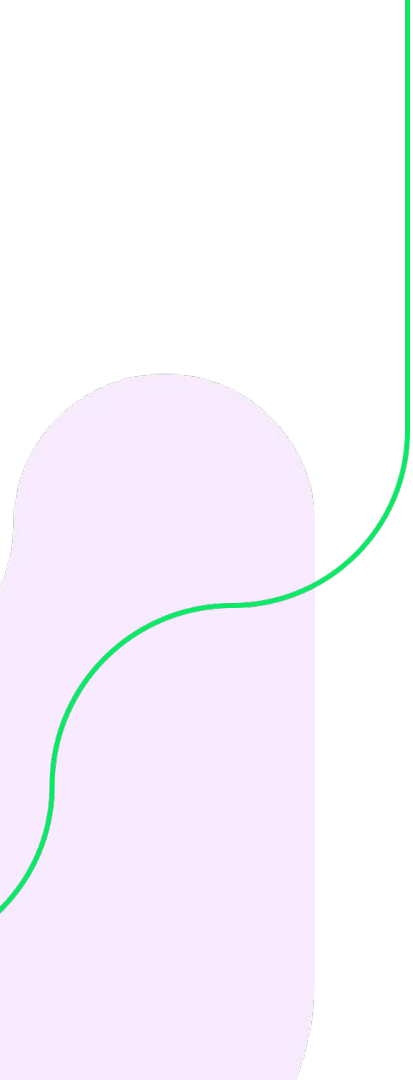
```
michael.lynn  27906 0.8 0.5 418566112 184592 ?? S   Wed07AM  74:39.24 /Applications/zoom.us.app/Contents/MacOS/zoom.us
```

```
michael.lynn  68596 0.8 0.3 444476624 130080 ?? S   Tue06PM  11:08.98 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/126.0.64
```

```
michael.lynn  79258 0.7 2.2 1600754960 837488 ?? S  10Jul24 110:46.85 /Applications/Visual Studio Code.app/Contents/Frameworks/Code Helper (Renderer).app/Contents/MacOS/Code
```

A woman with blonde hair, wearing a dark blazer over a pink top, is standing in a room that appears to be a library or office. She is looking towards the left. In the background, there is a wooden bookshelf filled with books and binders. To the left, there is a desk with a lamp and a black office chair. The text "THAT'S WHY EVERYBODY HATES THE LIBRARY." is overlaid in white, bold, sans-serif font at the bottom of the image.

**THAT'S WHY EVERYBODY
HATES THE LIBRARY.**



Get the titles of
books that
have an
average 5-star
rating





MongoDB Compass



SQL vs Aggregation Pipelines



Comparing SQL JOINS and aggregation

```
SELECT
    city,
    SUM(annual_spend) Total_Spend,
    AVG(annual_spend) Average_Spend,
    MAX(annual_spend) Max_Spend,
    COUNT(annual_spend) customers
FROM (
    SELECT t1.city, customer.annual_spend
    FROM customer
    LEFT JOIN (
        SELECT address.address_id, city.city,
               address.customer_id, address.location
        FROM address LEFT JOIN city
        ON address.city_id = city.city_id
        ) AS t1
    ON
    (customer.customer_id = t1.customer_id AND
     t1.location = "home")
) AS t2
GROUP BY city;
```





Comparing SQL JOINS and aggregation

```
db.customers.aggregate([
  {
    $unwind: "$address",
  },
  {
    $match: {"address.location": "home"}
  },
  {
    $group: {
      _id: "$address.city",
      totalSpend: {$sum: "$annualSpend"},
      averageSpend: {$avg: "$annualSpend"},
      maximumSpend: {$max: "$annualSpend"},
      customers: {$sum: 1}
    }
  }
])
```



Structure of an Aggregation Pipeline

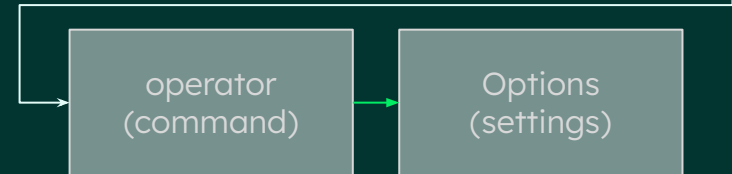


MongoDB Aggregation Syntax Structure

// Example MongoDB aggregation command

```
db.collection.aggregate([
  { $match: { age: { $gte: 25 } } },    // Stage 1: Match documents where age is greater than or equal to 25
  { $sort: { age: -1 } },              // Stage 2: Sort the results by age in descending order
  { $limit: 10 },                      // Stage 3: Limit the results to 10
  { $project: { name: 1, age: 1, _id: 0 } } // Stage 4: Include only the name and age fields in the results, exclude _id
]);
```

...



Agenda

 MongoDB Query API

 **Common Aggregation Stages**

 Building & Running Pipelines

 Wrap Up



Simple Pipelines

\$match

```
db.books.aggregate([
  {
    $match: {
      pages: 100
    }
  }
]);
```



\$project



```
db.books.aggregate([
  {
    $project: {
      title: 1,
      cover: 1
    }
  }
]);
```

\$limit

```
db.books.aggregate([
  {
    $limit: 5
  }
]);
```



\$count

```
db.books.aggregate([
  {
    $count: "numberOfBooks"
  }
]);
```



Using Arrays

Exact match



```
db.collection.aggregate([
  {
    $match: {
      tags: ['red', 'black']
    }
  }
]);
```

Match one element



```
db.collection.aggregate([
  {
    $match: {
      tags: 'red'
    }
  }
]);
```

Match multiple elements



```
db.collection.aggregate([
  {
    $match: {
      tags: {
        $all: ['red', 'black']
      }
    }
  }
]);
```



Joins

\$lookup



```
db.authors.aggregate([
  {
    $lookup: {
      from: "books",
      localField: "books",
      foreignField: "_id",
      as: "booksWritten"
    }
  }
])
```



Group results

\$group



```
db.books.aggregate([
  {
    $group: {
      _id: "$year",
      totalPages: {
        $sum: "$pages"
      }
    }
  }
])
```



Modifying results

\$addFields



```
db.books.aggregate([
  {
    $addFields: {
      readingTimeHours: {
        $divide: [
          { $multiply: ["$pages", 2] },
          60
        ]
      }
    }
  }
])
```



Exporting Data

\$out



```
db.books.aggregate([
  {$match: {pages: 100},
  {
    $out: {
      db: "reporting",
      coll: "authors"
    }
  }
])
```

Agenda

 MongoDB Query API

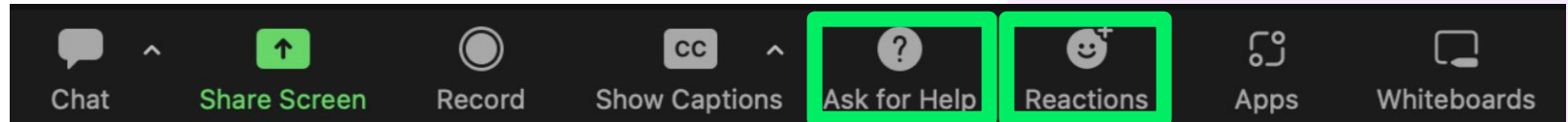
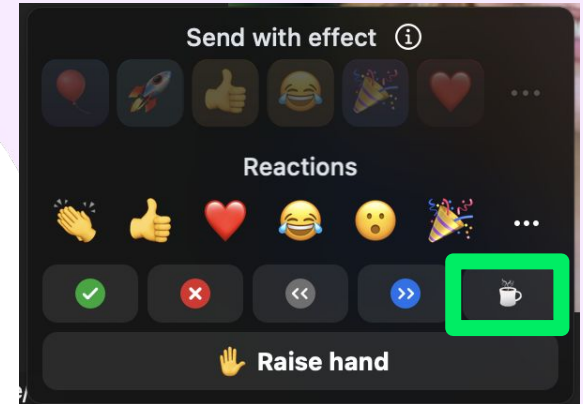
 Common Aggregation Stages

 **Building & Running Pipelines**

 Wrap Up

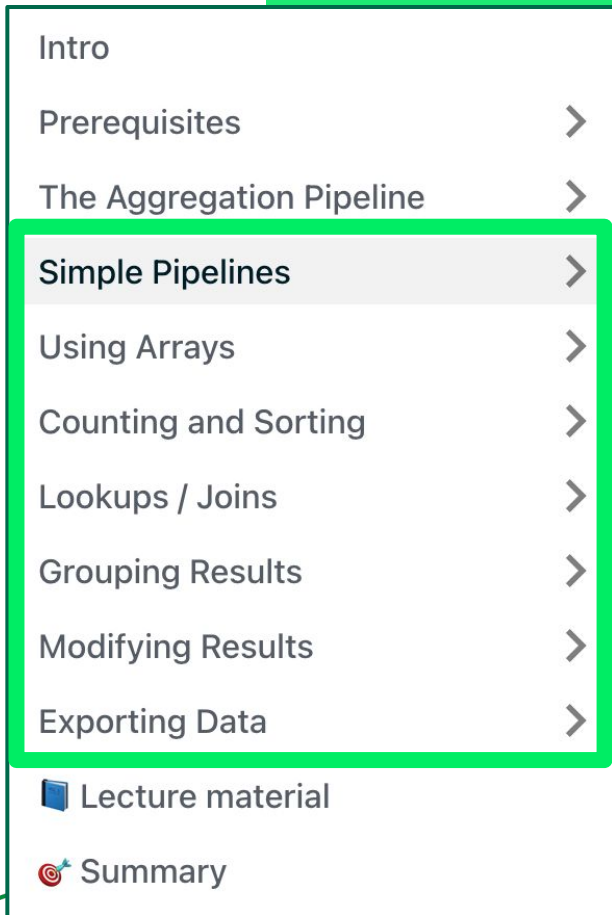
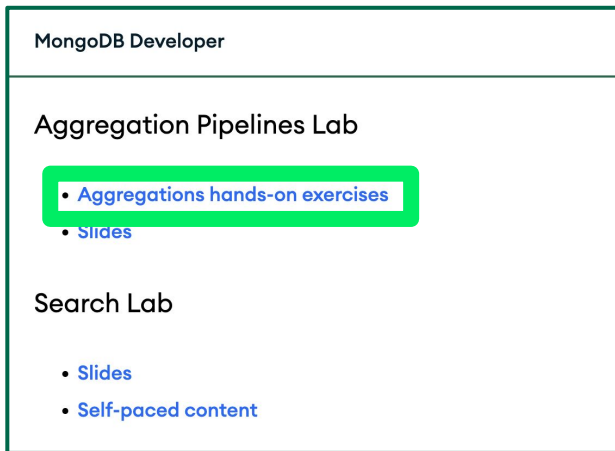


Zoom breakout rooms



At the end of this exercise, you will be able to:

- Run pipelines in Atlas
- Build your own pipelines using common stages
- Export data into a new collection



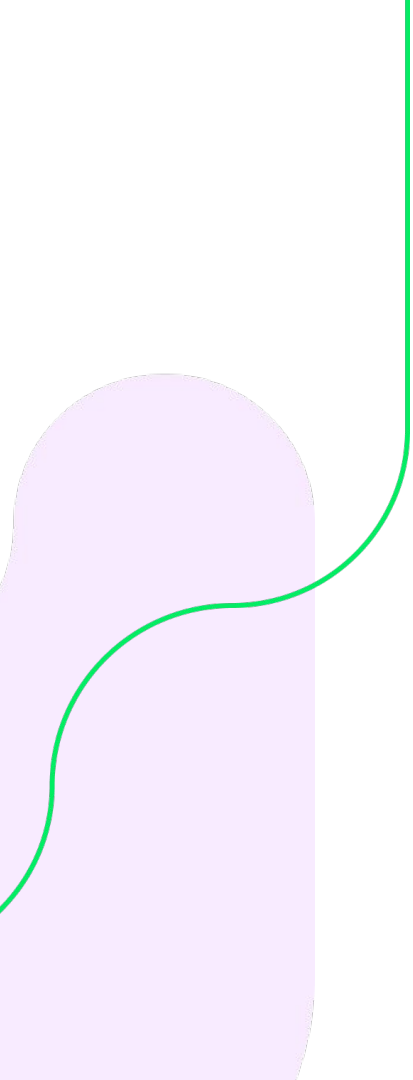
Agenda

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 **Wrap Up**



The **aggregation
framework** allows
you to transform
and analyze your
data in real time.



Aggregation features

A feature-rich framework for data transformation and analytics

Pipeline stages

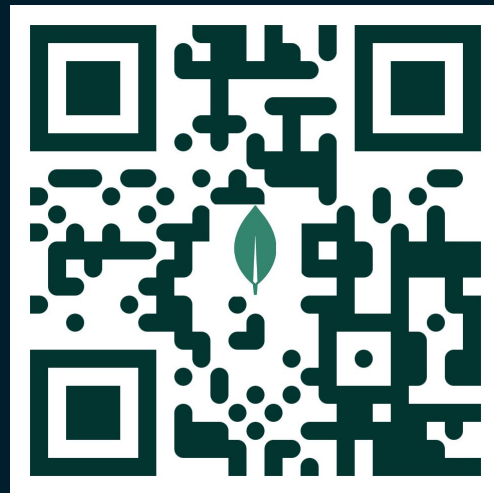
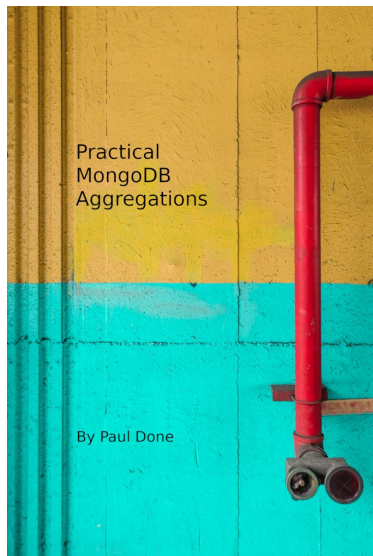
- \$match
- \$group
- \$facet
- \$geoNear
- \$graphLookup
- \$lookup
- \$merge
- \$project
- \$search
- \$sort
- \$setWindowFields
- \$unionWith
- \$unwind
- ...and more

Operators

- Mathematical
 - \$add, \$abs, \$subtract, \$multiply, \$divide, \$log, \$log10, \$stdDevPop, \$stdDevSam, \$avg, \$sqrt, \$pow, \$sum, \$zip, \$convert, \$round, etc.
- Array
 - \$push, \$reduce, \$reverseArray, \$addToSet, \$arrayElemAt, \$slice, etc.
- Conditionals
 - \$and, \$or, \$eq, \$lt, \$lte, \$gt, \$gte, \$cmp, \$cond, \$switch, \$in, etc.
- Temporal
 - Window functions
 - \$dateAdd, \$dateDiff, \$dateSubtract, \$dateTrunc
 - \$dateFromParts, \$dateToParts, \$dateFromString, \$dateToString, \$dayOfMonth, \$isoWeek, \$minute, \$month, \$year, etc.
- String
 - \$toUpper, \$toLower, \$substr, \$strcasecmp, \$concat, \$split, etc.
- Literals
 - \$exp, \$let, \$literal, \$map, \$type, etc.
- Regex
 - \$regexFind, \$regexMatch, etc
- Trigonometry
 - \$sin, \$cos, \$degreesToRadians, etc.
- Custom aggregation expressions

E-BOOK

Practical MongoDB Aggregations

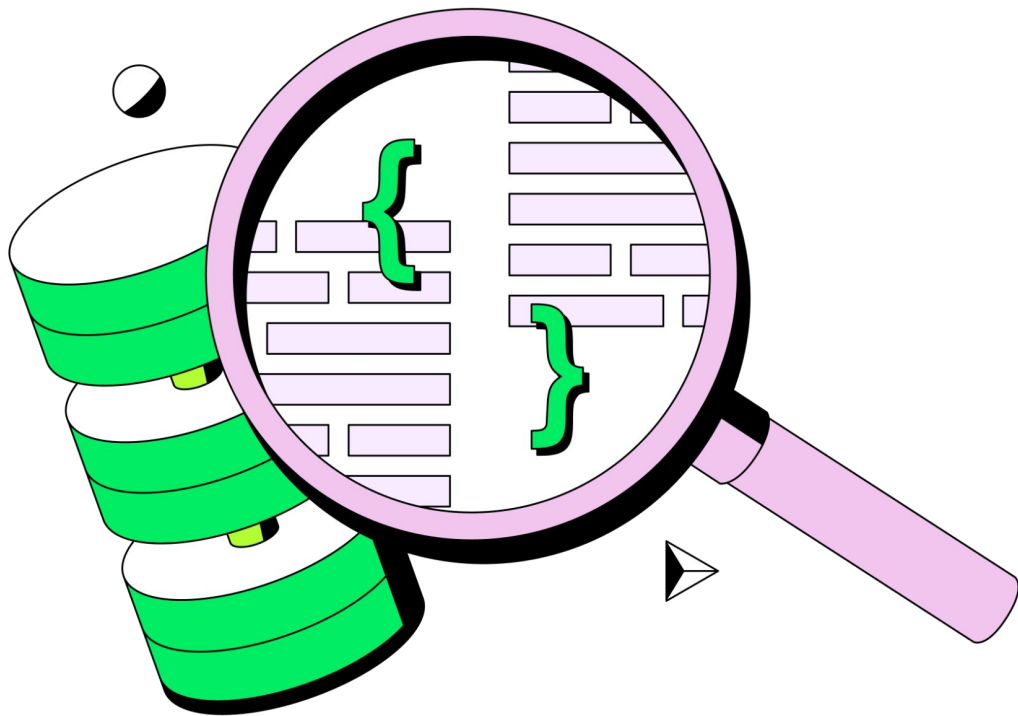


mongodb.link/agg-ebook



The MongoDB Query API is rich

- Expressive queries
- Aggregations
- Join collections
- Graph queries
- Geospatial queries
- Full text search
- Vector search



Session survey

Any
questions?

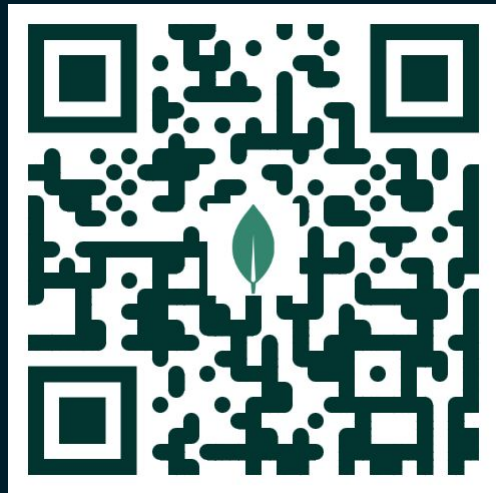
Thank you!



DESIGN REVIEW

Request a Session!

Scan the QR code to request a design review slot this afternoon for you and your team to answer all your application development questions!



mdb.link/devday-design-review